

# Robot Learning: From Fundamentals to Foundation Models

## Lecture 10: Generalist Robot Policies

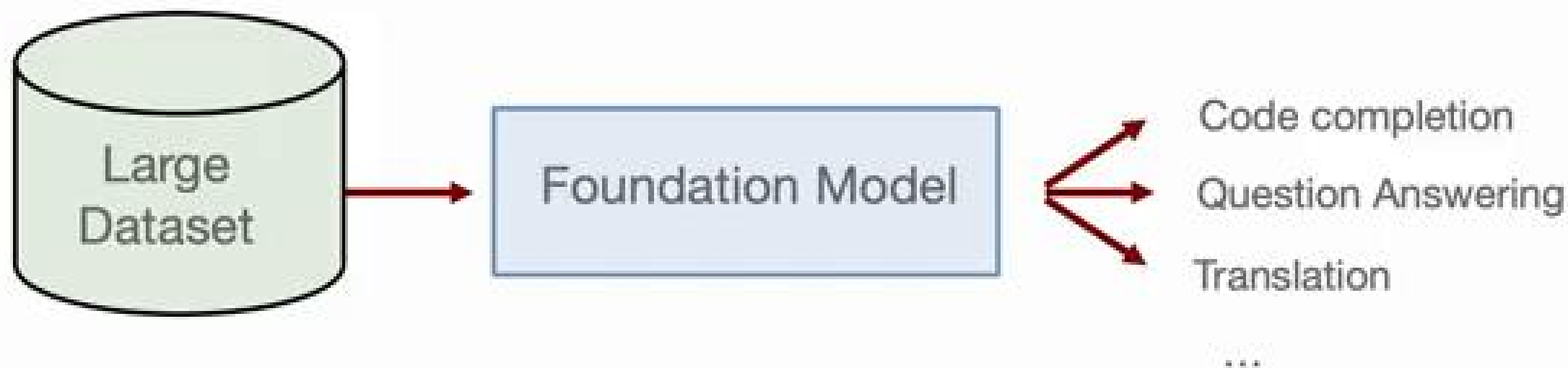
Oier Mees

**ETH** zürich

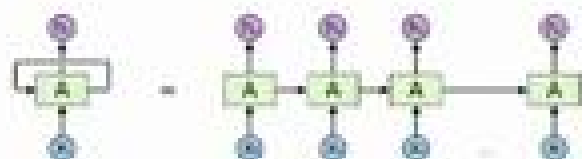


27.04.2026

# Foundation Models in NLP



# Foundation Models > Specialist Models



Translation: Seq2Seq, GNMT...



Object Detection: R-CNN, Fast-RCNN...



Image Captioning: DenseCap, Show & Tell..



Vision-Language Models: ViLBERT, Flamingo, BLIP, GPT-4V...

# Ingredients for Generalist Robot Policies



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## Large Datasets

Robot Data is Scarce

Collecting Data  
requires Human  
Supervision



# Ingredients for Generalist Robot Policies

## Large Datasets

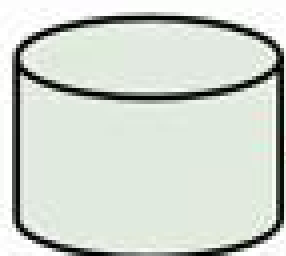
Robot Data is **Scarce**

Collecting Data  
requires **Human  
Supervision**

## Large Models

Robot Data is  
**Multimodal &  
Heterogeneous**

Robots need **High  
Frequency Control**



# Ingredients for Generalist Robot Policies

## Large Datasets

Robot Data is **Scarce**

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## Large Models

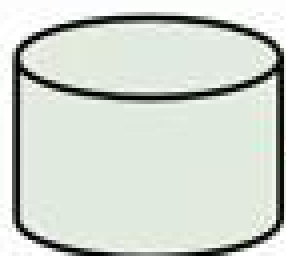
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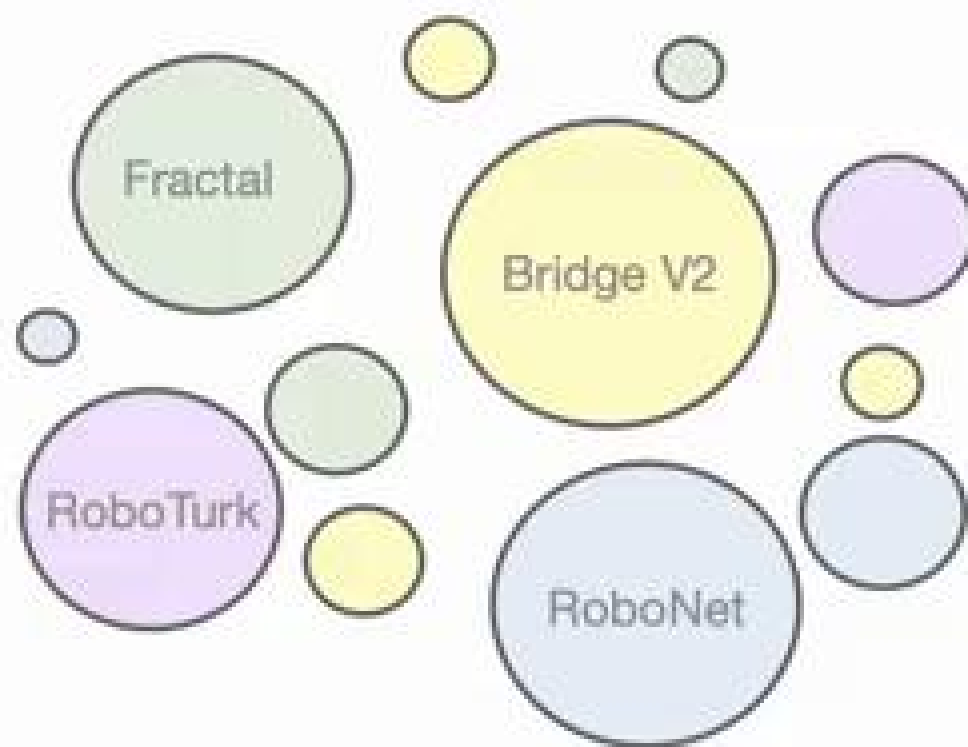
## Scalable Evaluation

Robot Evals are  
**Tedious & Expensive**

**Difficult to reproduce**

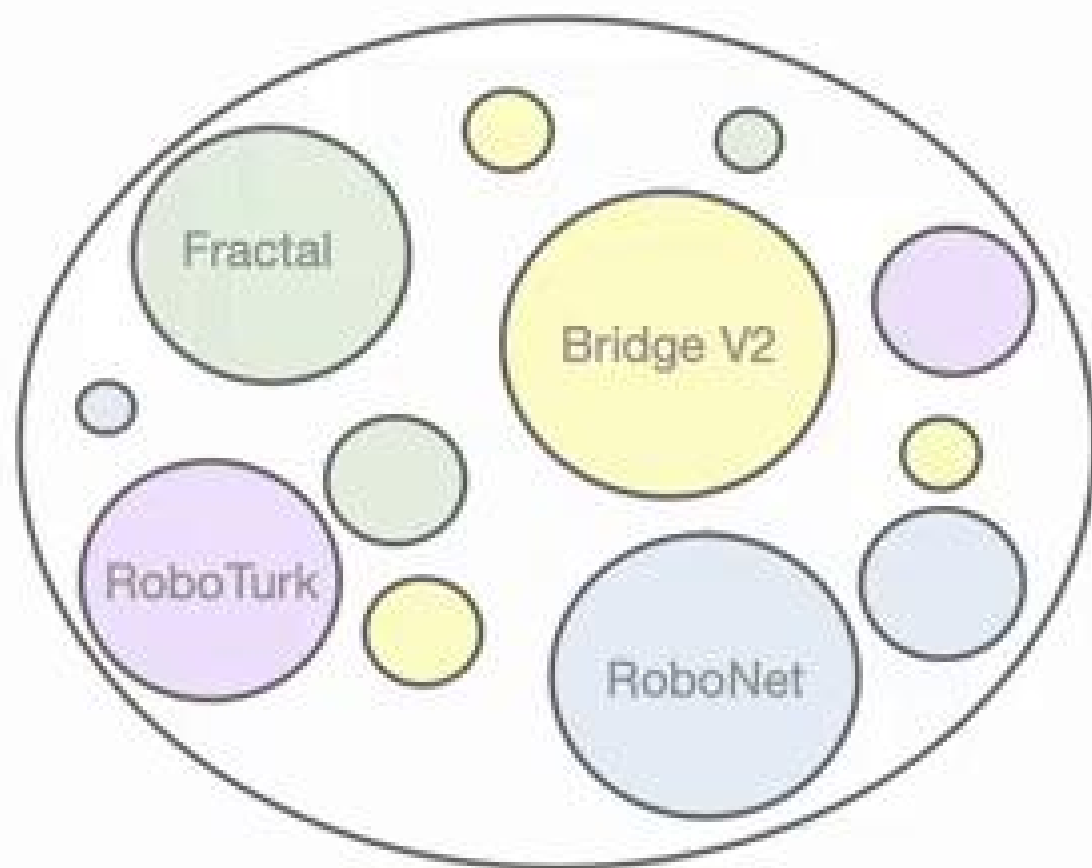


# Key Idea: Leverage **Existing** Robot Datasets



# Key Idea: Leverage **Existing** Robot Datasets

**Aggregate** existing data into **common format**



# Large Robot Data: Open X-Embodiment



-  **1M+** Real Robot Episodes
-  **22** Robot Embodiments
-  **34** Research Labs
-  **300+** Scenes

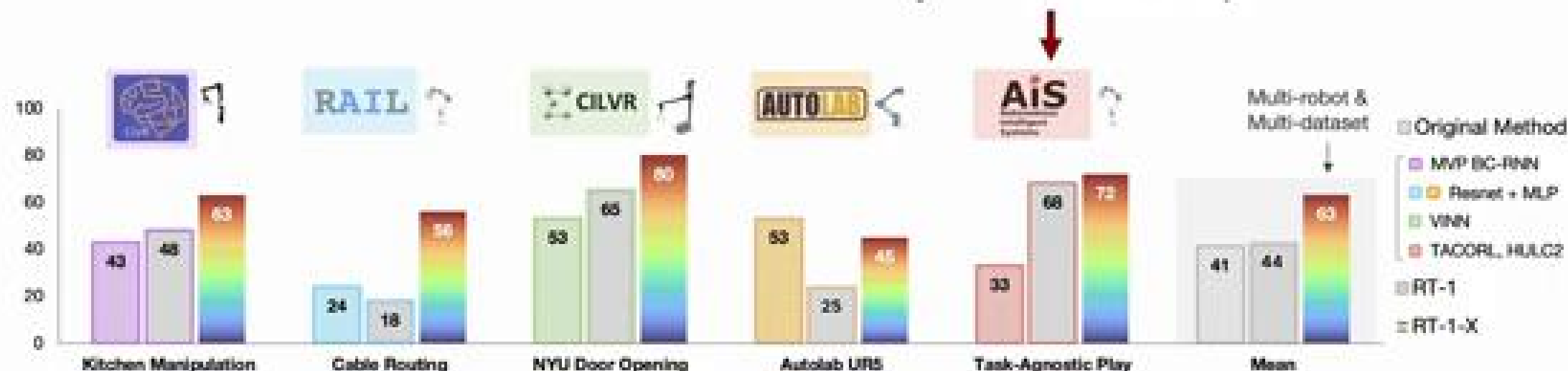
Open X-Embodiment: Robotic Learning Datasets and RT-X Models

Open X-Embodiment Collaboration, ..., Mees et al., ICRA, 2024. **Best Conference Paper Award (out of 1765 papers)**

# Sparks of a Generalist Robot Policy



My PhD robot setup



Open X-Embodiment: Robotic Learning Datasets and RT-X Models

Open X-Embodiment Collaboration, ..., Mees et al., ICRA, 2024. **Best Conference Paper Award (out of 1765 papers)**

# Ingredients for Generalist Robot Policies

## Large Datasets

Leverage Existing Robot Data



## Large Models

Robot Data is Multimodal & Heterogeneous

Robots need High Frequency Control

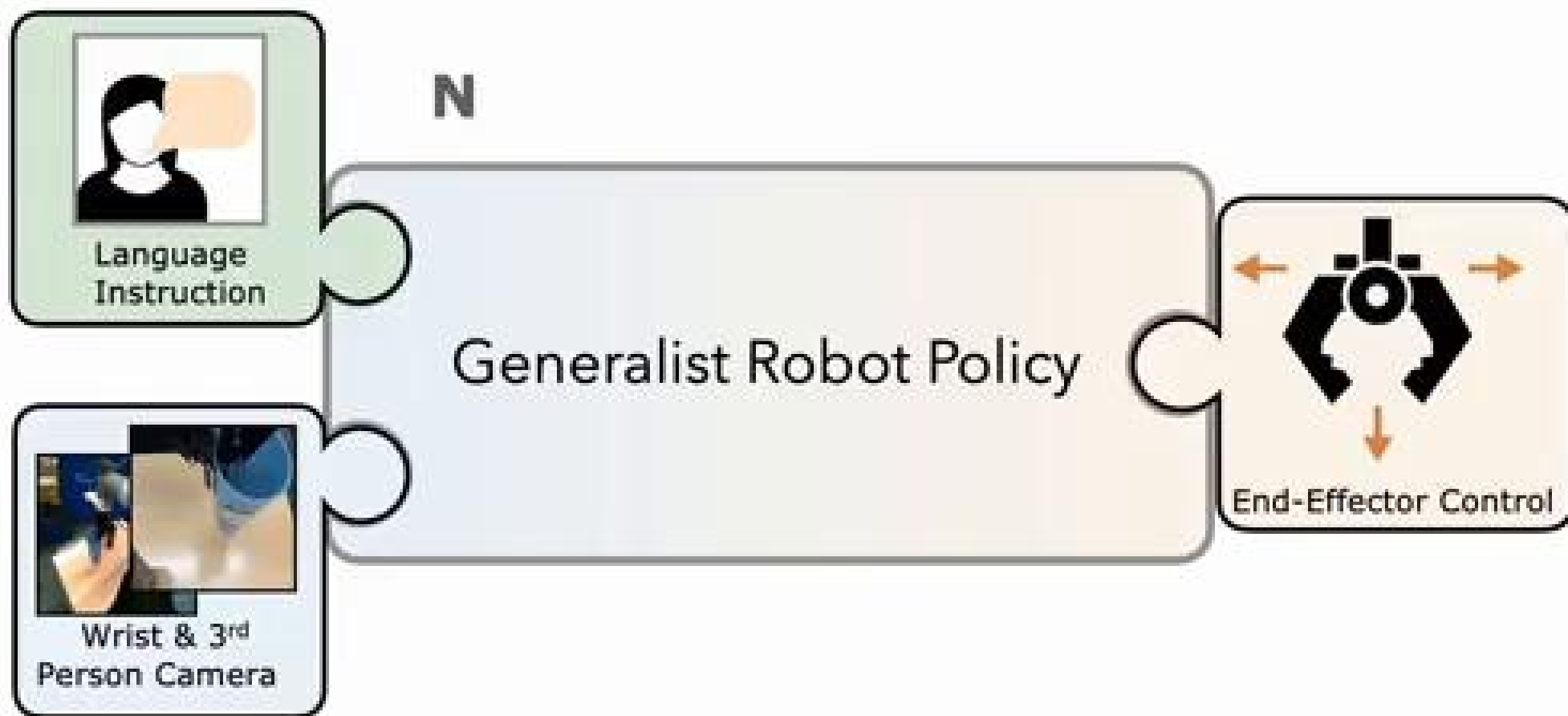
Robot Foundation Model

## Scalable Evaluation

Robot Evals are Tedious & Expensive

Difficult to reproduce







**New Observation**



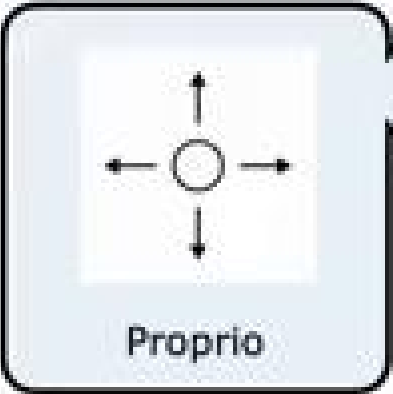
Generalist Robot Policy

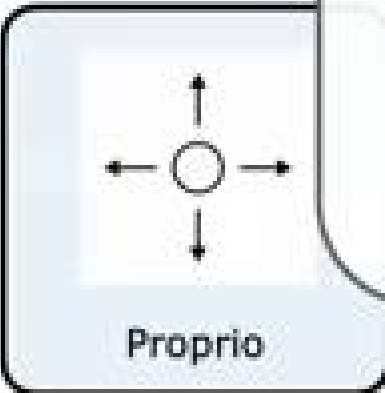




# New Action Space

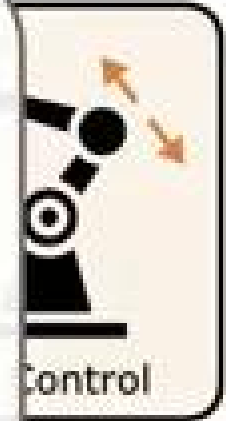
Generalist Robot Policy





## New Action Space

- Robot Data is **Heterogeneous**
- **Diverse** Sensors
- **Diverse** Actuators
- **Diverse** Control Frequencies





"Pick up the  
spoon"

Policy



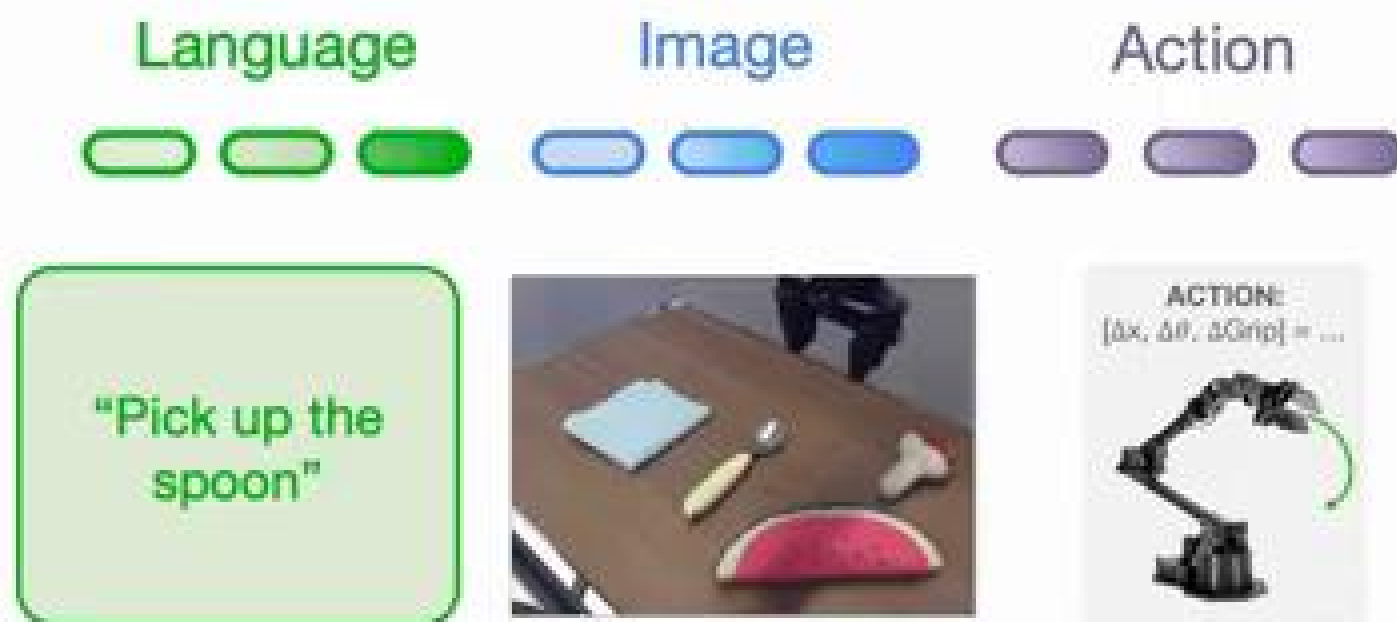


"Caption the scene"

Vision-Language Model

"The picture shows the Statue of Liberty in NY"

# Key: Robotics as Multimodal Sequence Modeling





# Octo Architecture



Octo: An Open-Source Generalist Policy

Octo Model Team, Ghosh\*, Walke\*, Pertsch\*, Black\*, Mees\* et al., RSS, 2024.



# Octo Architecture

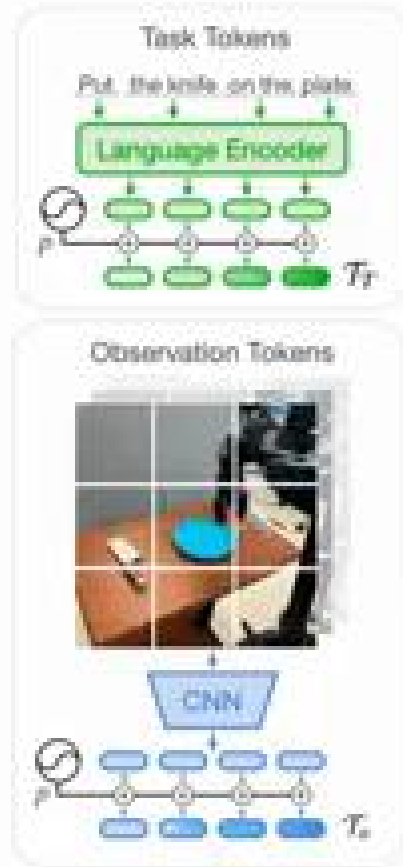


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# Octo Architecture

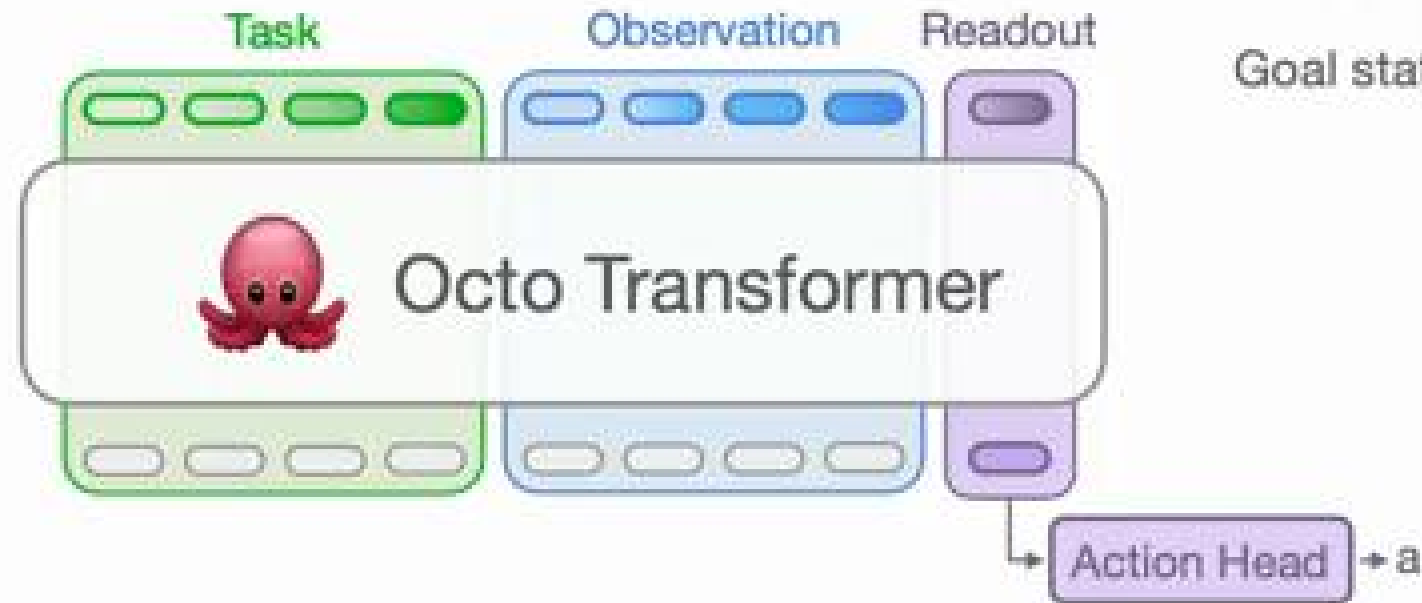
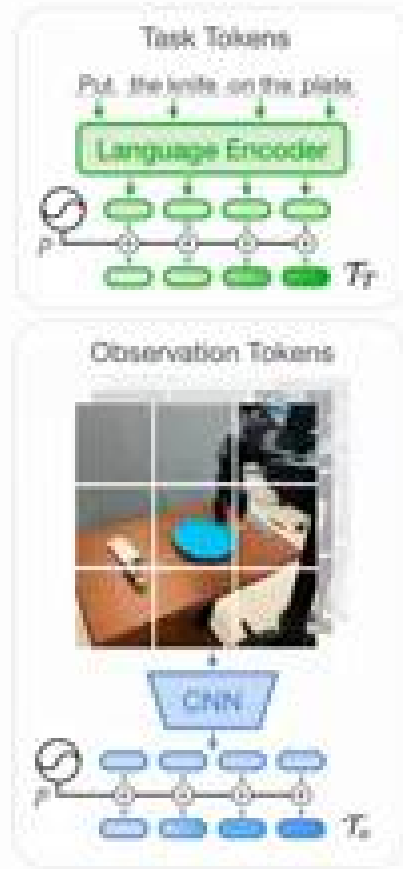


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# Octo Architecture



goal image, language



$$\pi_{\theta}(a_t | s_t, s_g)$$

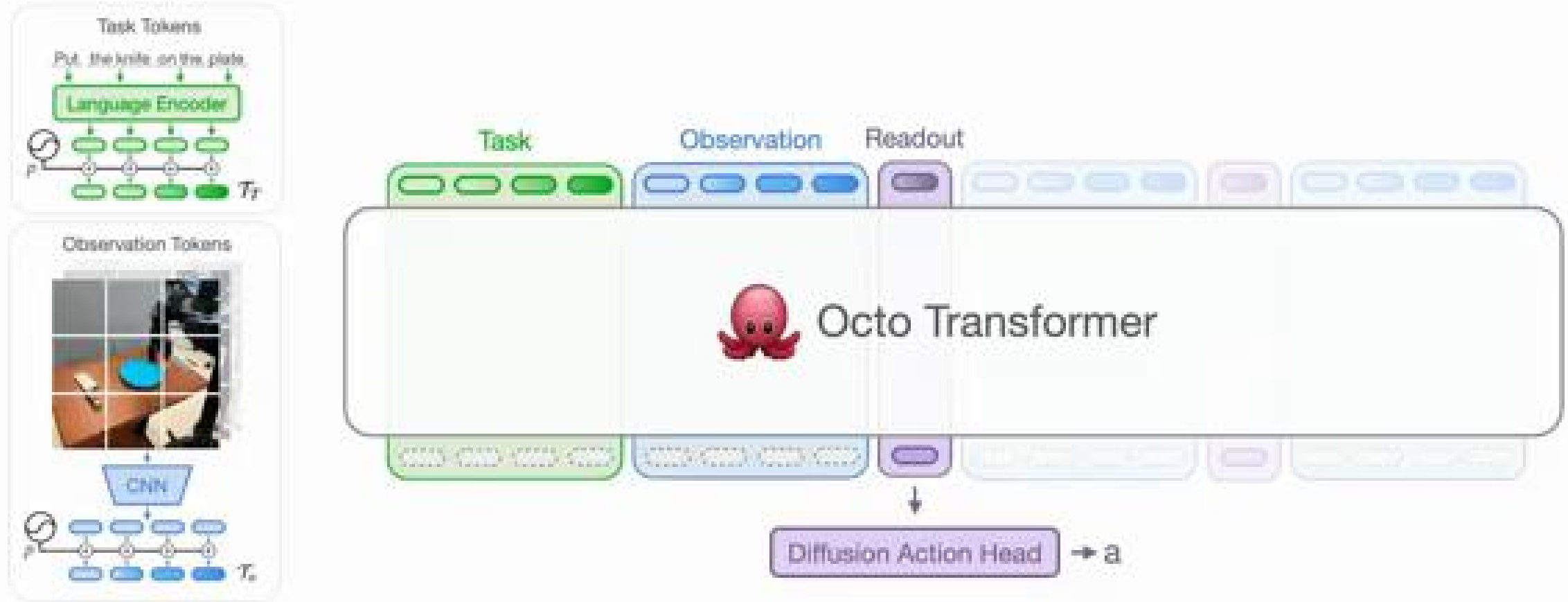
Goal state conditioned BC

Octo: An Open-Source Generalist Policy

Octo Model Team, Ghosh\*, Walke\*, Pertsch\*, Black\*, Mees\* et al., RSS, 2024.



# Octo Architecture



Octo: An Open-Source Generalist Policy

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# Octo Design Decisions

- Train everything from scratch, most params in Transformer
- Align gripper action across OXE datasets

## Gripper action conventions in OXE

Typical motion (open → closed to grasp → open to release), four dataset conventions

### Absolute, +1 open / 0 closed

Taco, Austin Sailor, Austin Sirius, NYU Franka Play



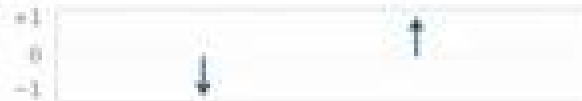
### Absolute, 0 open / 1 closed (inverted)

Roboturk, Viola, Stanford Hydra, BC-Z



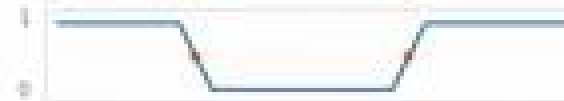
### Relative deltas, +1 opening / -1 closing

RT-1, Kuka, jaco Play, Berkeley Autolab UAS



### Continuous with transition ramps

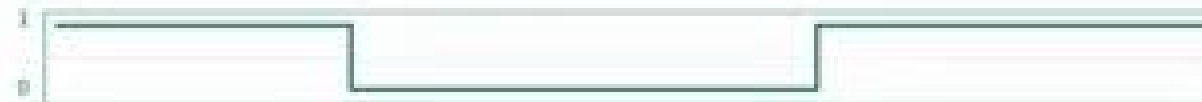
Bridge — intermediate values during opening/closing



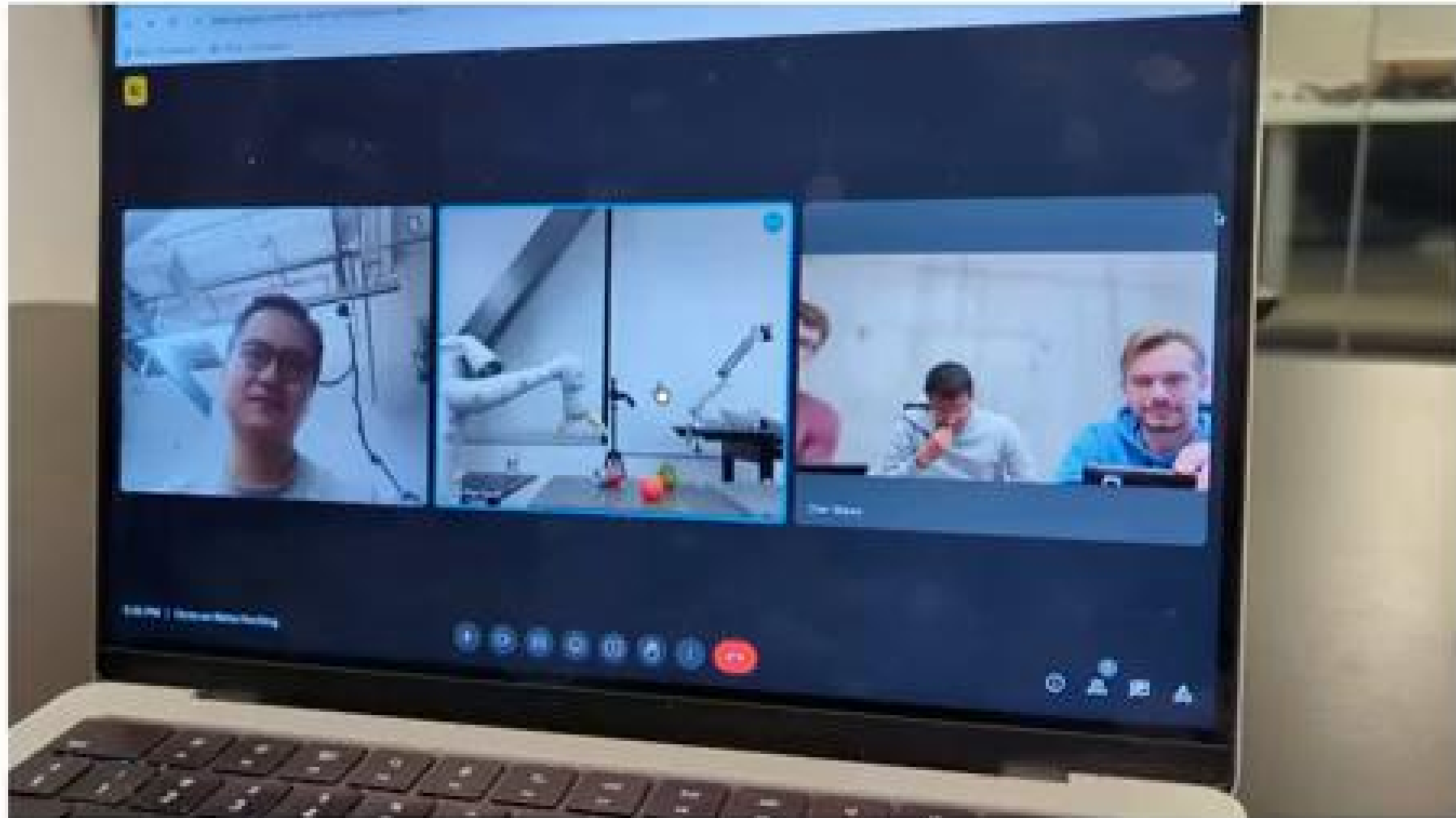
orange points: analog values during transitions

Octo aligns all four to a single convention

### Target: absolute, +1 open / 0 closed



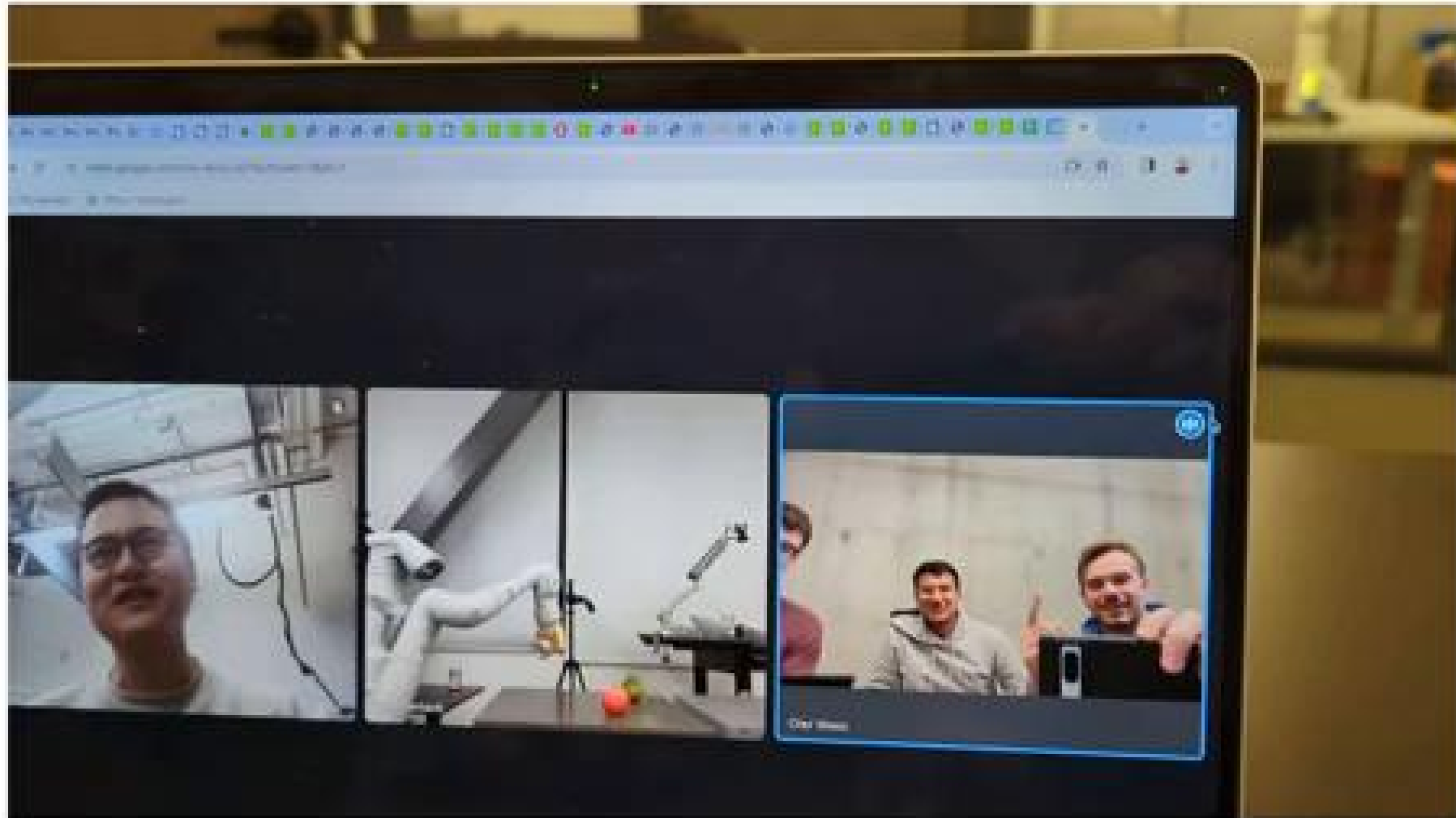
# The Eureka Moment



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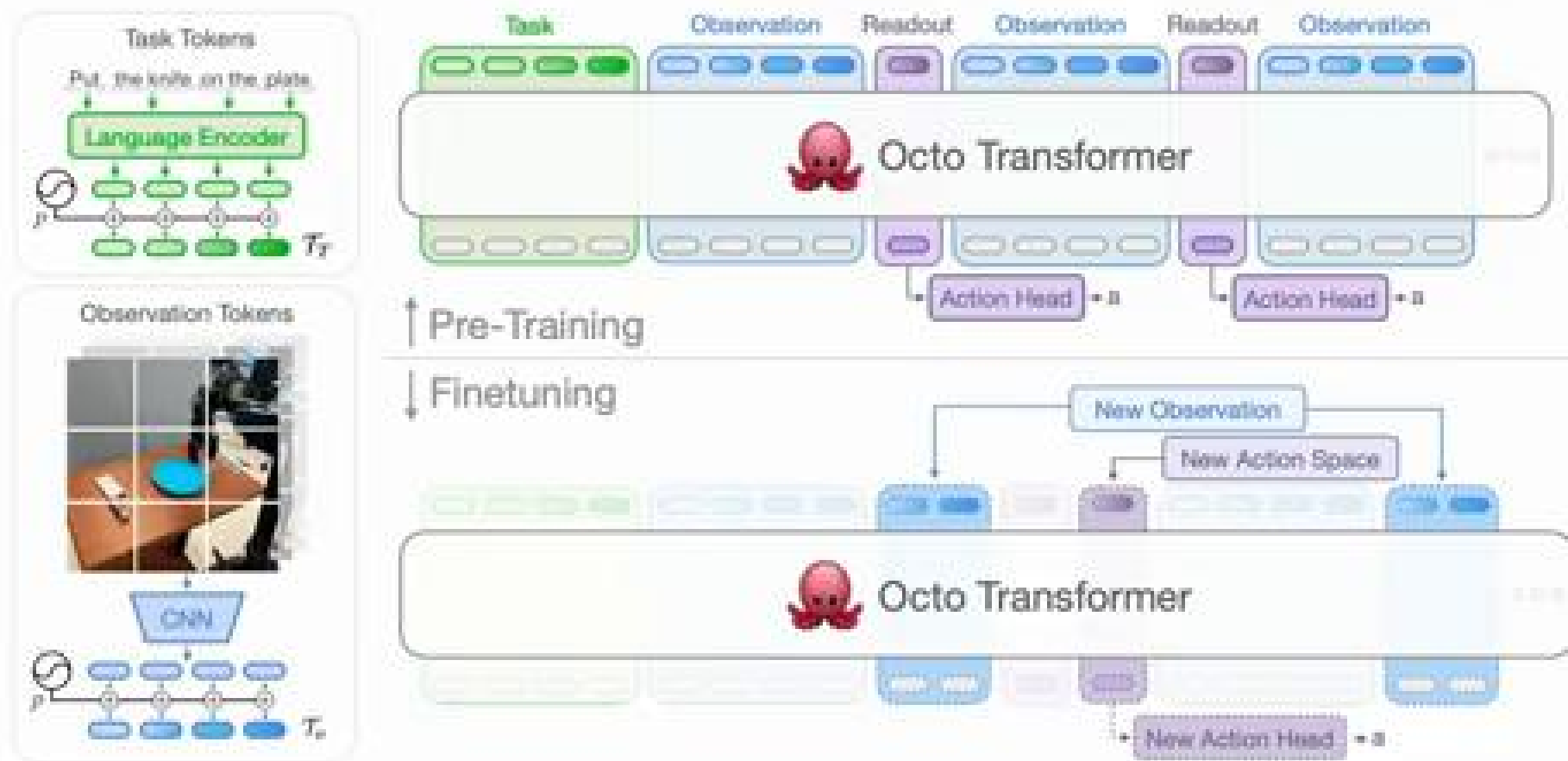


**How To Adapt To New Obs/Action Spaces?**



# Octo Finetuning

**Adaptable** pre-trained representation for **finetuning**



Octo: An Open-Source Generalist Policy

Octo Model Team, Ghosh\*, Walke\*, Pertsch\*, Black\*, Mees\* et al., RSS, 2024.

# Adoption of Octo

- Key: RFMs learn **better representations** for **transfer**
- Researchers finetune to their robot with 50 demos

```
from octo import OctoModel  
  
model = OctoModel.load_pretrained("hf://rail-berkeley/octo-base")
```



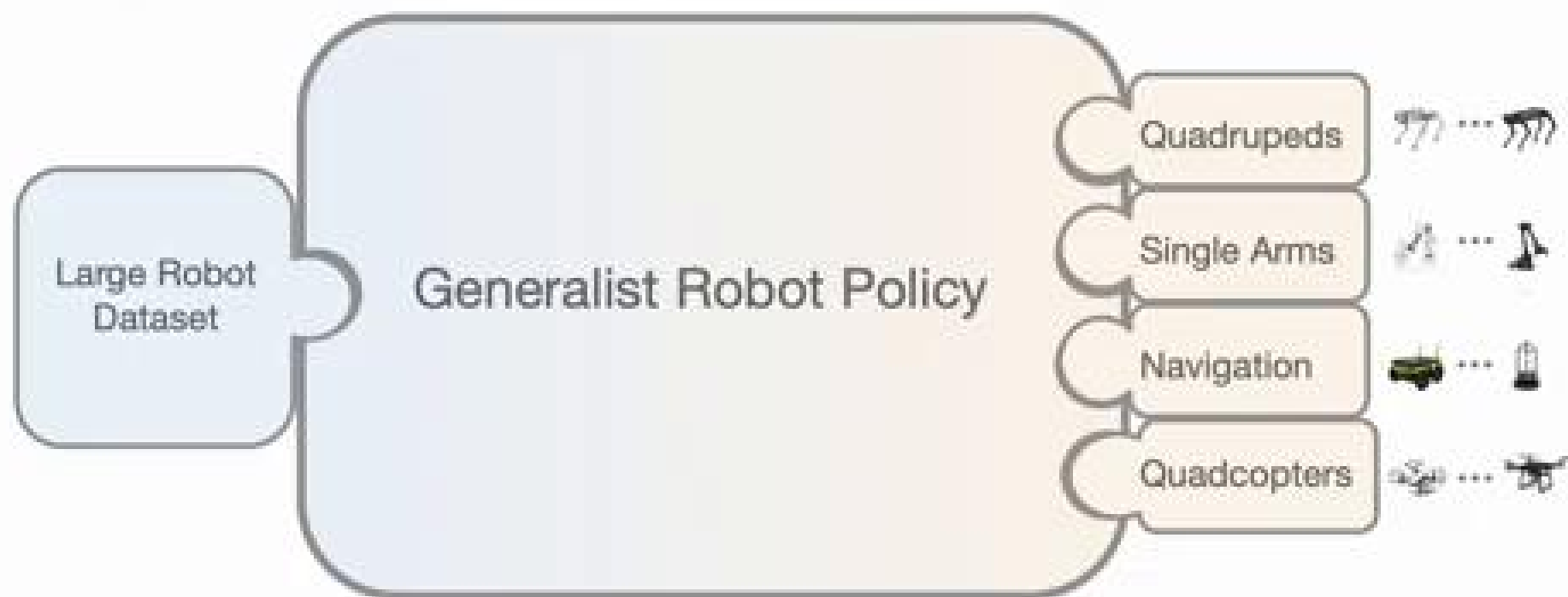
Source: Peter Mitrano via Twitter



Source: Tokyo Robotics

# Cross-Embodiment Learning

- How “generalist” is a single arm manipulation policy?
- Can we **learn from more robot embodiments**?
- **Transfer** knowledge across embodiments



# One Policy for Manipulation, Navigation, Locomotion & Aviation



Scaling Cross-Embodied Learning: One Policy for Manipulation, Navigation, Locomotion and Aviation

Doshi\*, Walke\*, Mees, Dasari and Levine. CoRL 2024. **Oral, top %4 of 670 papers.**

# One Policy for Manipulation, Navigation, Locomotion & Aviation

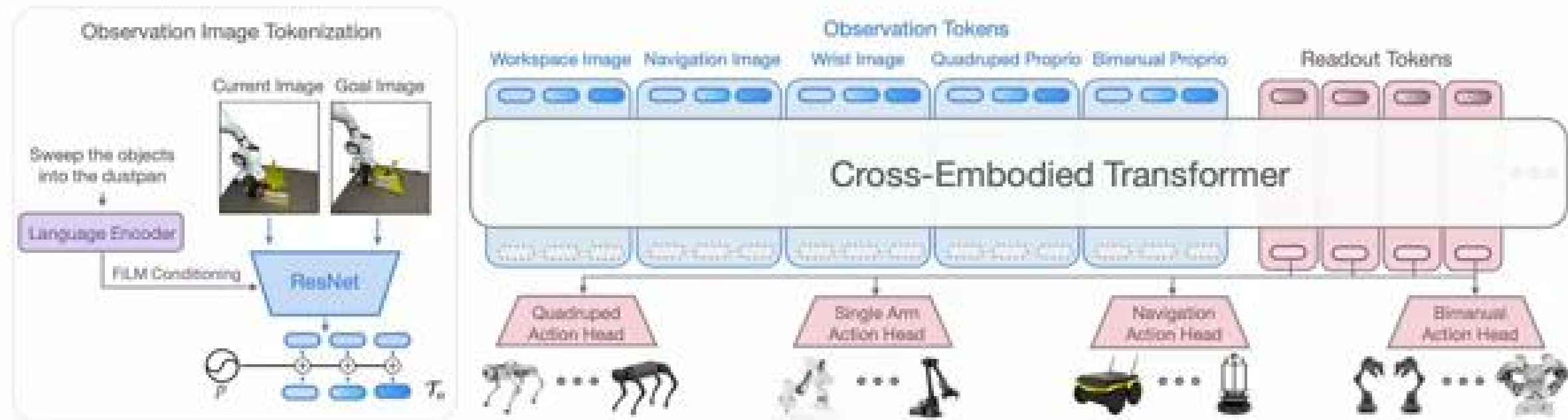


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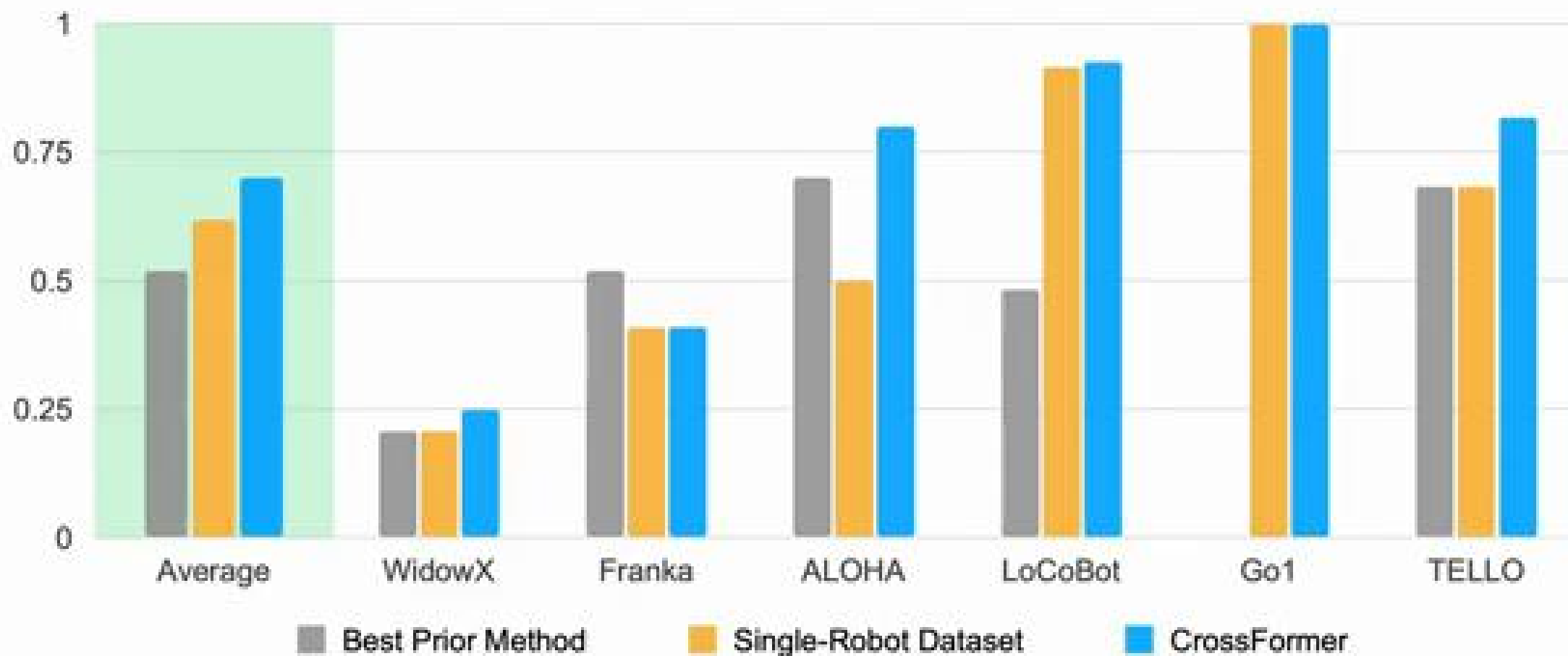
# CrossFormer Model

- No action space alignment required
- Can consume data from *any* robot embodiment
- Maximize parameter sharing across embodiments



# Quantitative Results

Key: matches and **outperforms specialist policies**

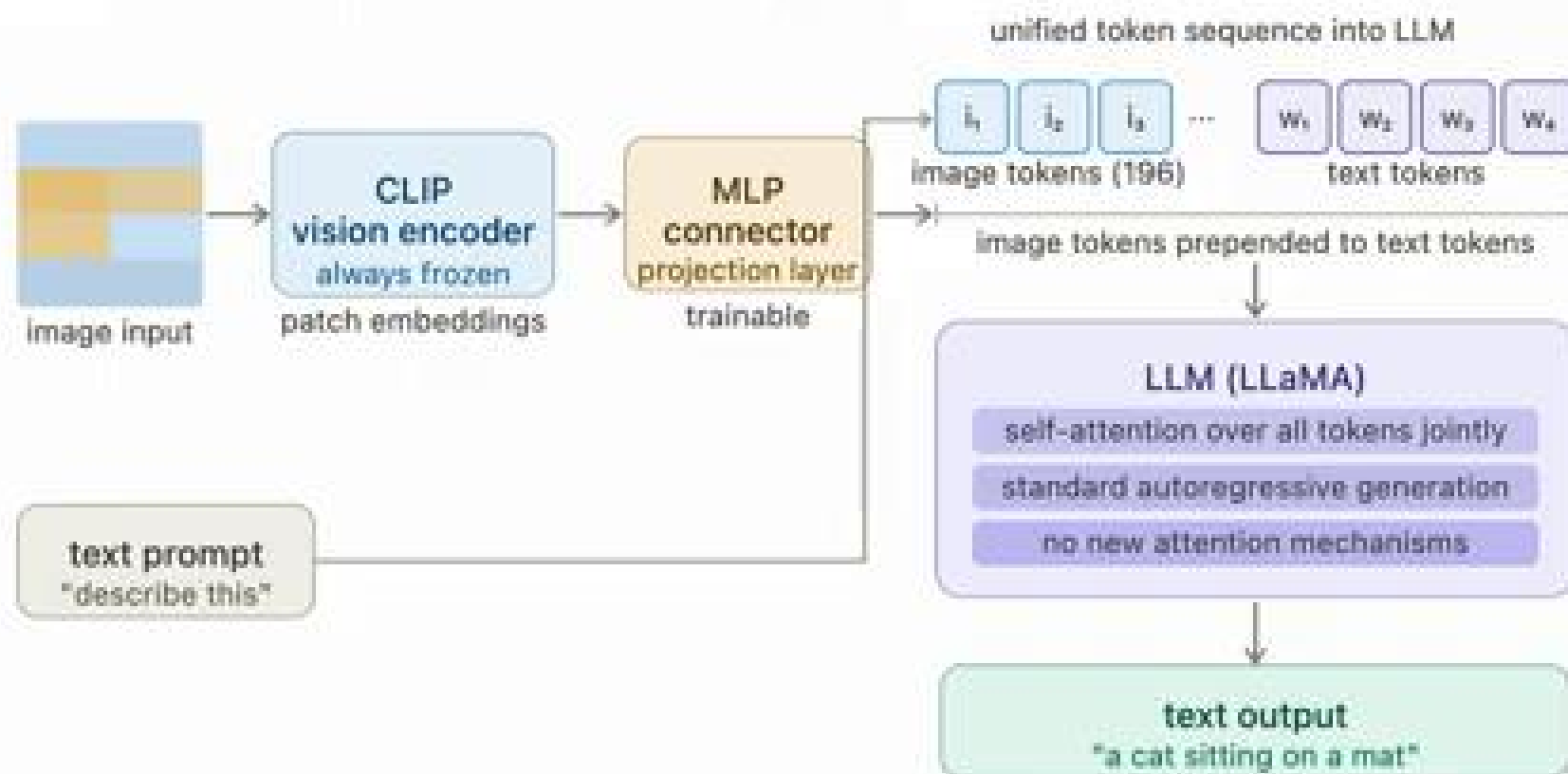


Everything so far trains on robot data **from scratch**

**Do internet-scale priors like VLMs transfer to control?**

# Recap Llava

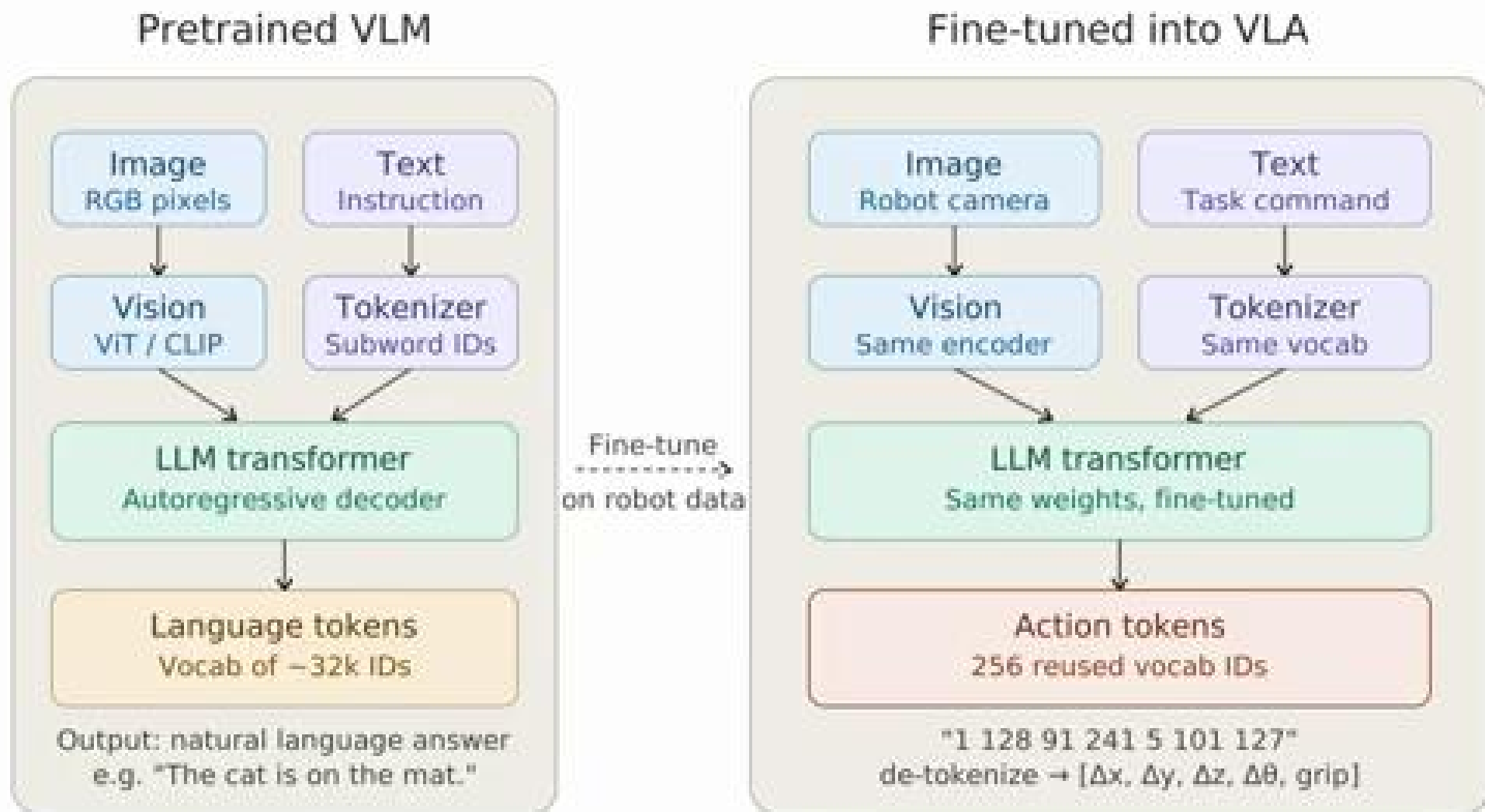
- Prepends image tokens to the text sequence, LLM processes everything in single unified self-attention pass



LLaVA treats images as another token sequence the LLM can attend to

**Can we treat robot actions the same way?**

# VLM $\longrightarrow$ Vision-Language-Action Model (VLA)

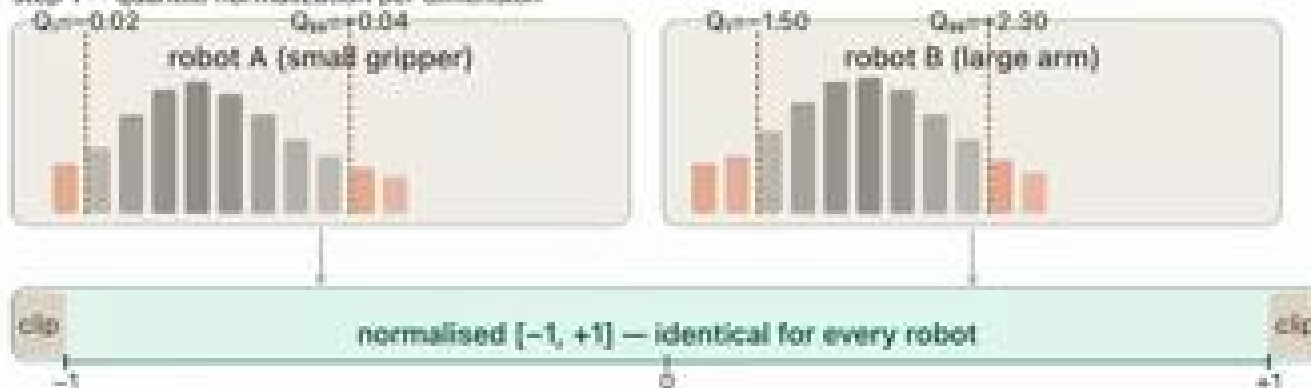


# Recap: Robot Action Tokenization for VLAs

## Per-Dimension, Per-Timestep Binning

Quantile bounds prevent outliers from expanding the discretization range and wasting bin resolution

step 1 — quantile normalization per dimension



step 2 — divide  $[-1, +1]$  uniformly into 256 bins · bin width =  $2/256 = 0.0078$



step 3 — N-dim action at timestep t → N discrete tokens

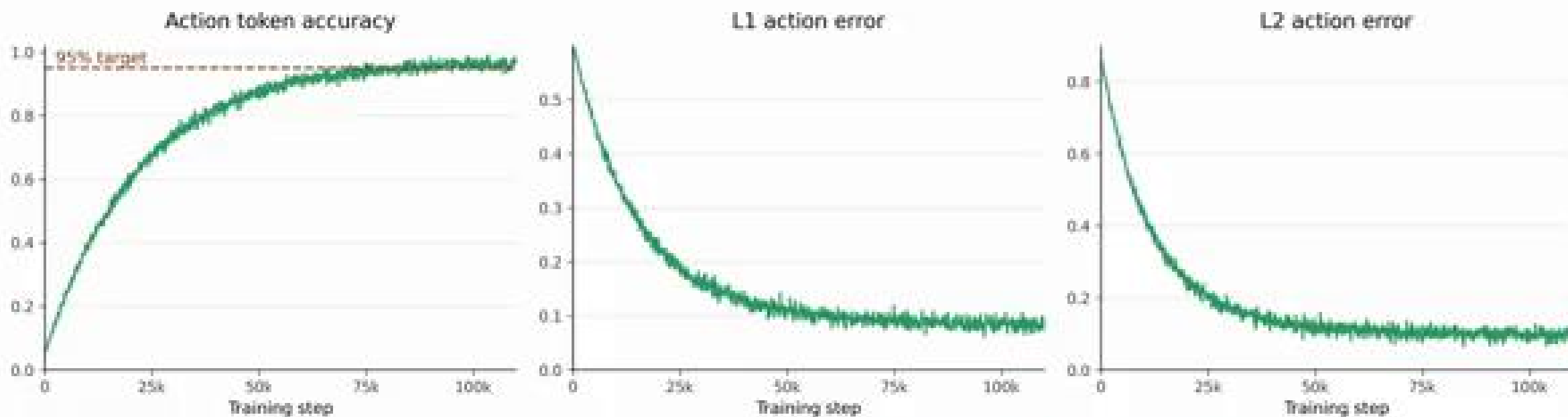


step 4 — inject action tokens into pre-trained language model vocabulary · train with next-token prediction



# Tips & Tricks: Convergence

- How do I know my VLA training has converged?



Did we pick the exact action bin?

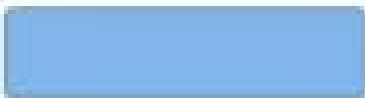
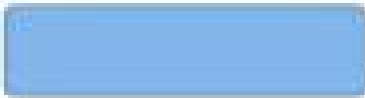
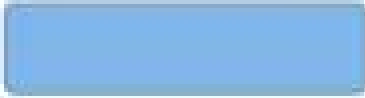
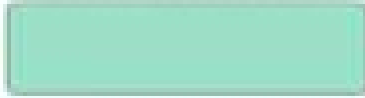
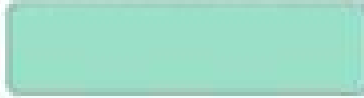
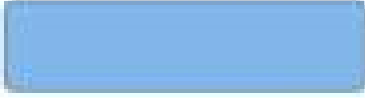
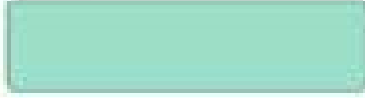
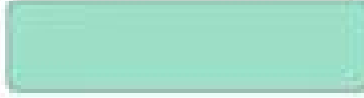
How far off are the predicted continuous actions (after detokenization) from ground-truth actions?

# Tips & Tricks: Batching Heterogeneous Datasets

- Datasets in OXE have different observation spaces
- Padding & Masking:
  - Easiest to implement
  - Half your batch will be full of zeros, wasted FLOPS

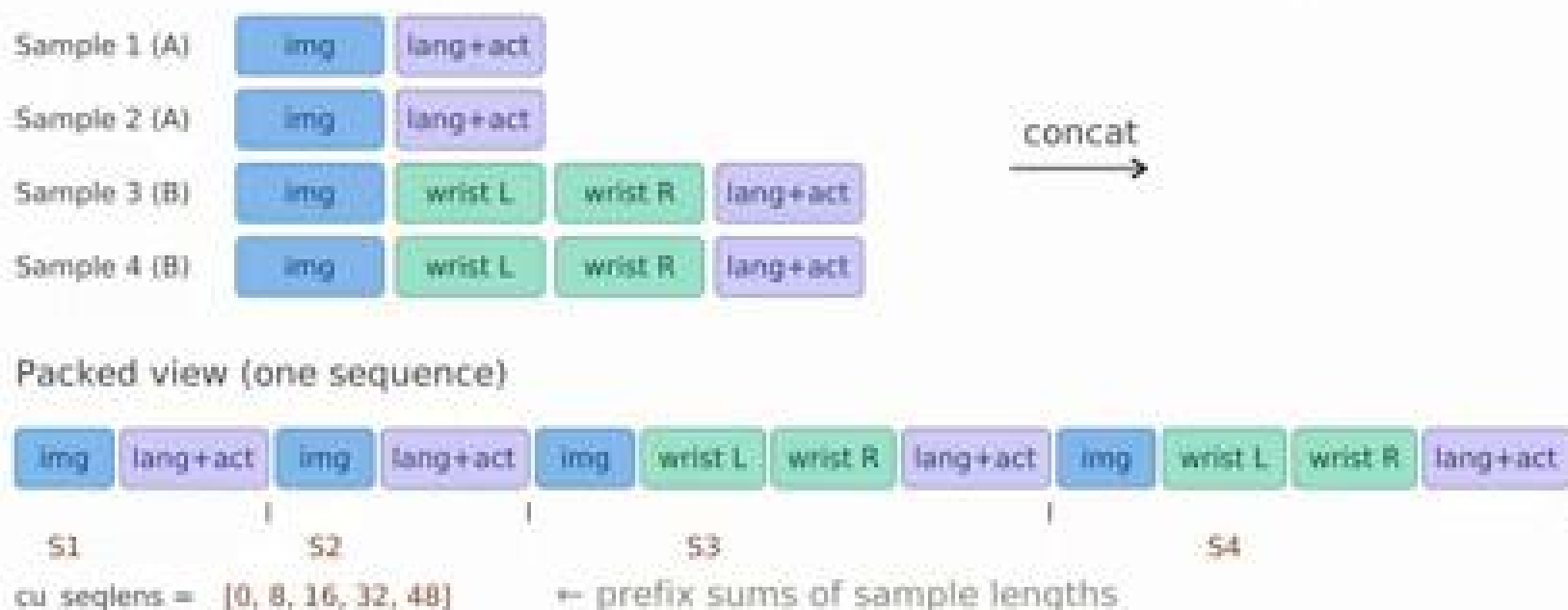
Dataset A: 1 third-person camera

Dataset B: 1 third-person + 2 wrist cameras

|          |           | 3rd-person Cam                                                                       | Wrist L Cam                                                                           | Wrist R Cam                                                                           | Lang + actions                                                                        |
|----------|-----------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Sample 1 | Dataset A |   |   |   |   |
| Sample 2 | Dataset A |  |  |  |  |
| Sample 3 | Dataset B |  |  |  |  |
| Sample 4 | Dataset B |  |  |  |  |

# Tips & Tricks: Batching Heterogeneous Datasets

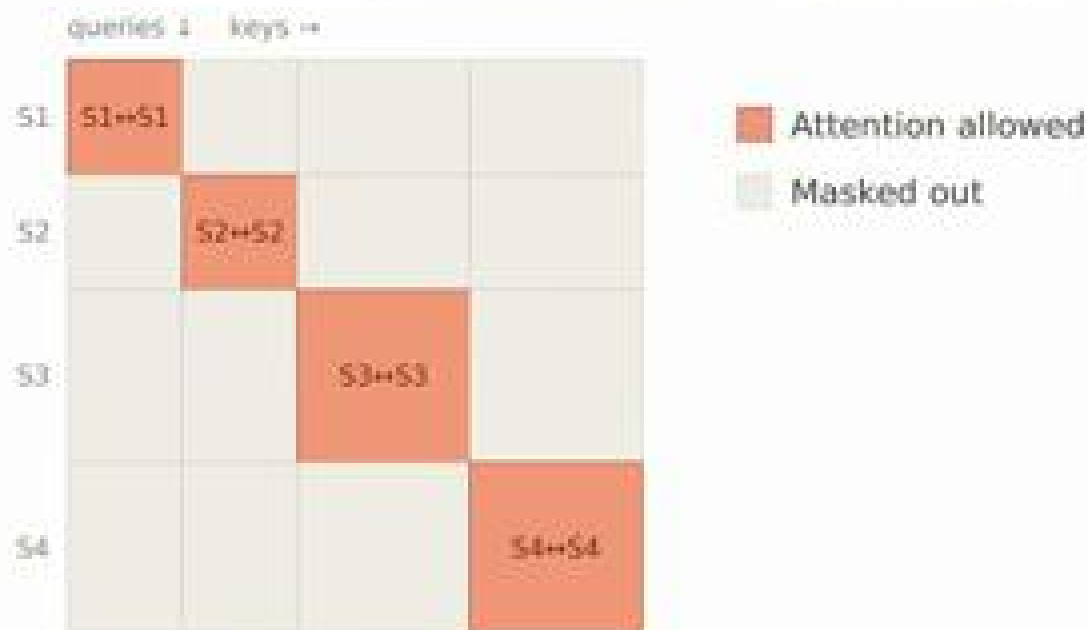
- Datasets in OXE have different observation spaces
- Sequence Packing:
  - Maximize GPU Throughput
  - Impl Complexity: reset positional encodings, block-diagonal att.



# Tips & Tricks: Batching Heterogeneous Datasets

## 1. Attention must be block-diagonal

Tokens in sample  $i$  can only attend to other tokens in sample  $i$ .  
Without this mask, sample 1 would attend to sample 3's images.

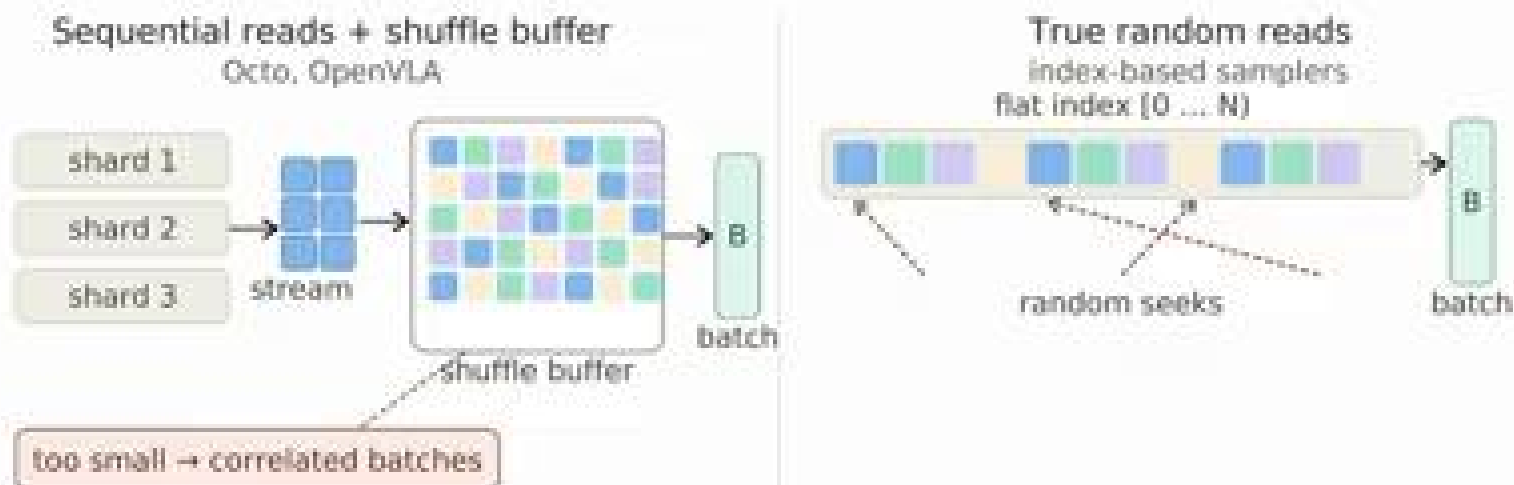


## 2. Positional encodings reset at each boundary

Token 0 of every sample gets position 0 — not its absolute offset in the packed sequence.  
Otherwise sample 4 would see "position 24+" as out-of-distribution input.

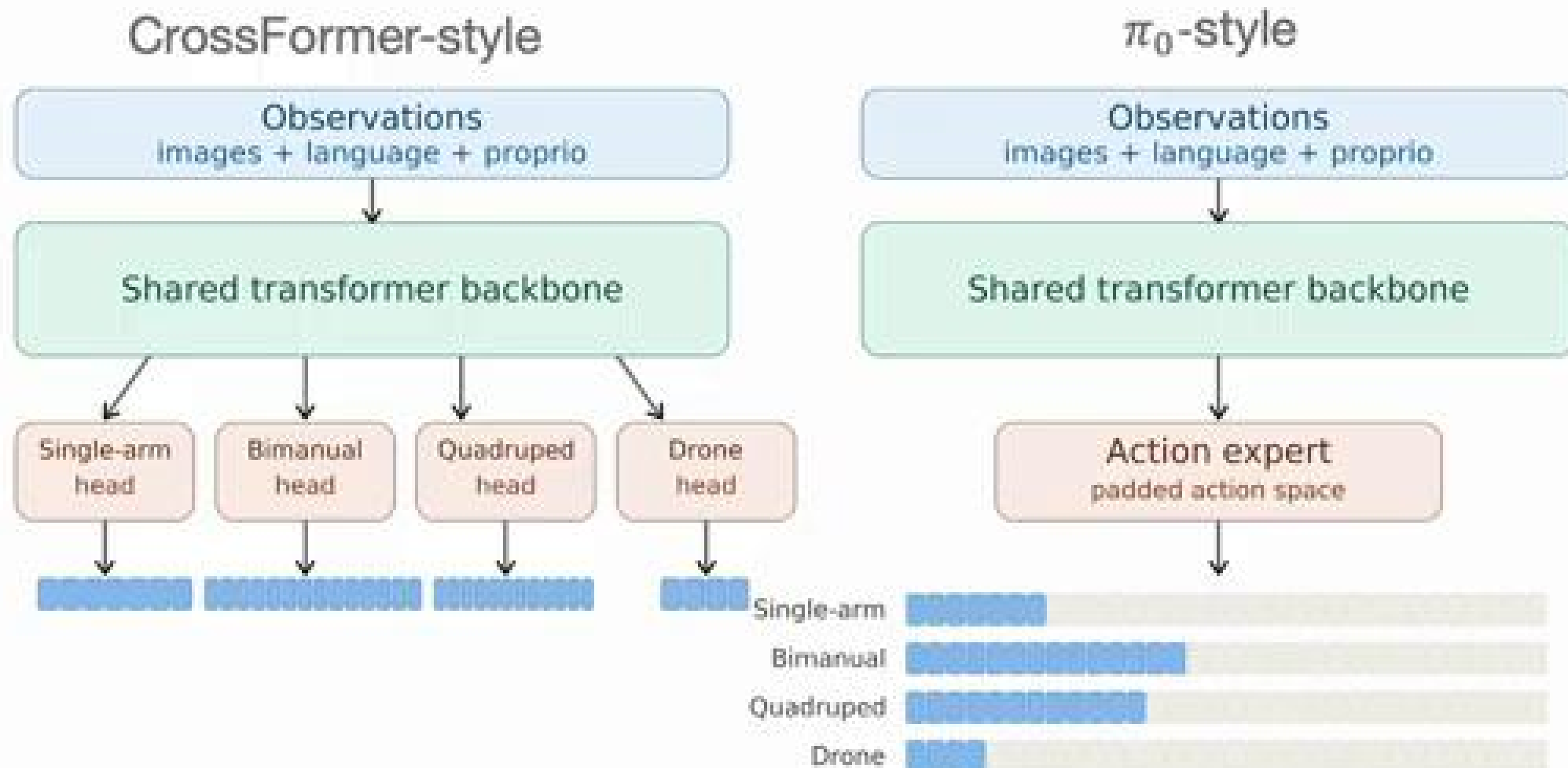


# Tips & Tricks: Dataloading



|              | Sequential + buffer                                                                                                                                                                                                                                                                                                                                                                                | True random reads |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| I/O pattern  | sequential reads                                                                                                                                                                                                                                                                                                                                                                                   | random seeks      |
| throughput   | very high                                                                                                                                                                                                                                                                                                                                                                                          | lower             |
| randomness   | approximate                                                                                                                                                                                                                                                                                                                                                                                        | exact             |
| memory       | large buffer in RAM                                                                                                                                                                                                                                                                                                                                                                                | index only        |
| obs. history | free — adjacent                                                                                                                                                                                                                                                                                                                                                                                    | extra seeks       |
|              | <div style="display: flex; align-items: center;"> <div style="width: 10px; height: 10px; background-color: #c8e6c9; border: 1px solid #000; margin-right: 5px;"></div> advantage         </div> <div style="display: flex; align-items: center;"> <div style="width: 10px; height: 10px; background-color: #ffcdd2; border: 1px solid #000; margin-right: 5px;"></div> disadvantage         </div> |                   |

# Tips & Tricks: Cross-Embodiment Heads

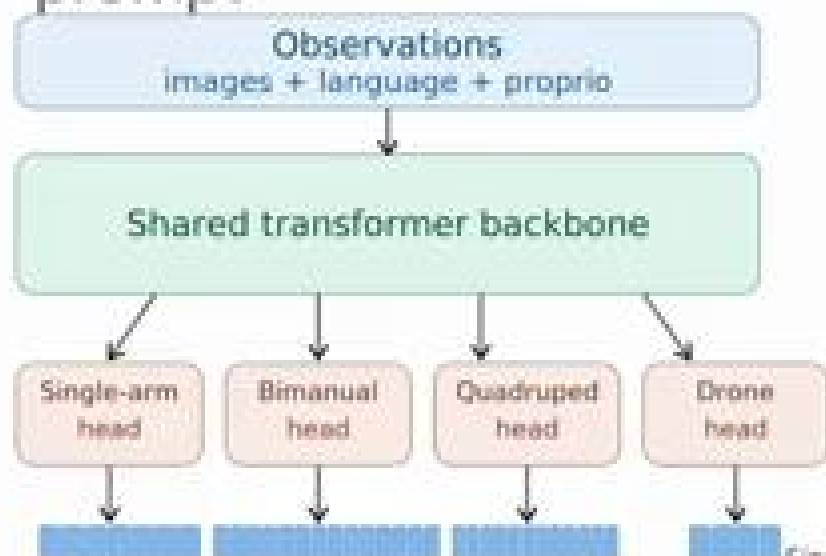


**How does the VLA know which embodiment to produce actions for at test time?**

# Cross-Embodiment Heads

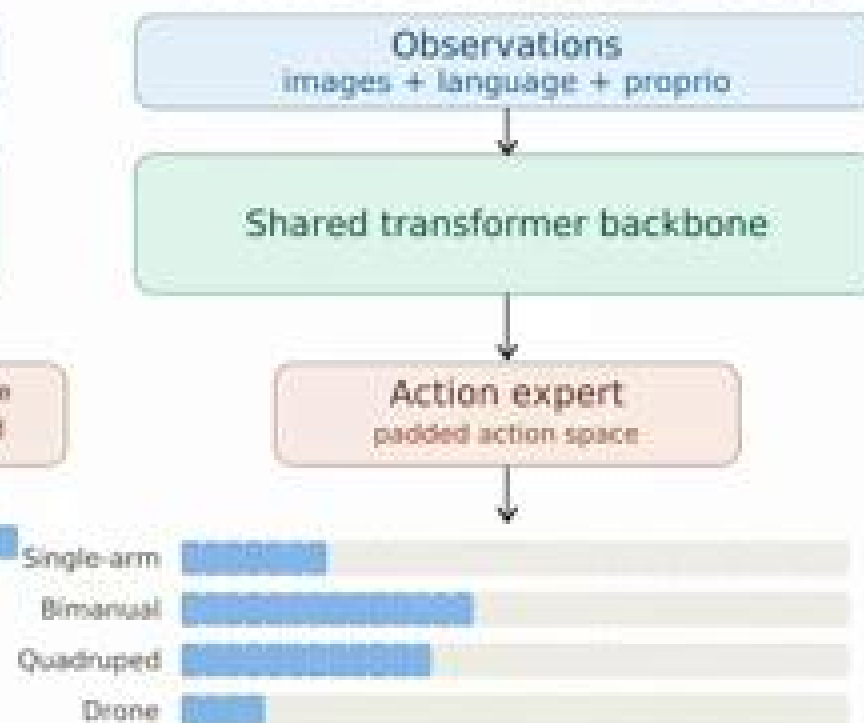
## CrossFormer-style

User specifies head in prompt



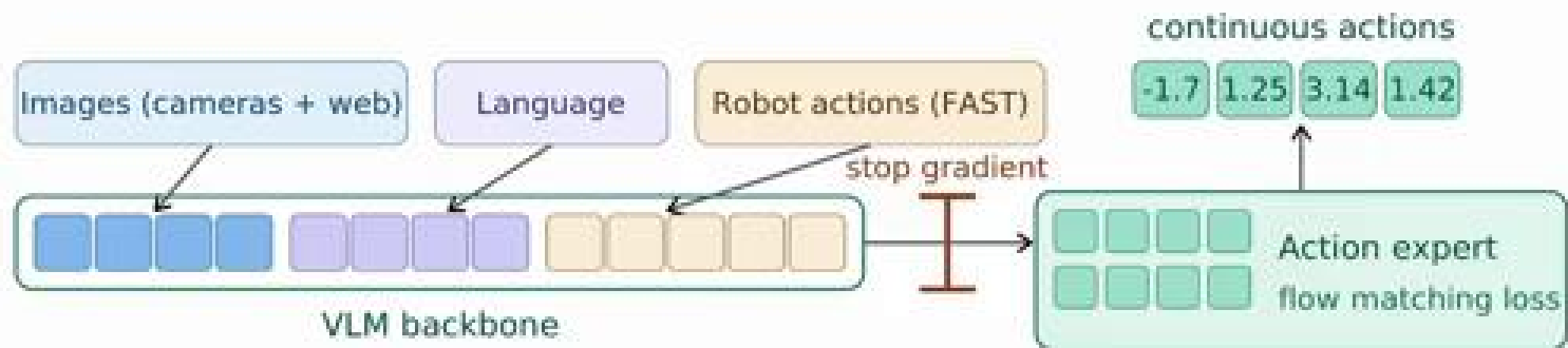
## $\pi_0$ -style

Model infers embodiment from obs, proprio & task



# Current VLA Recipe

- Next token prediction for VLM: FAST robot actions + Web
- Single action expert: flow matching + stop gradient to VLM



■ image tokens   ■ language tokens (NTP loss)   ■ FAST action tokens (NTP loss)   ■ continuous actions

✗ stop gradient — action expert cannot update backbone

# Ingredients for Generalist Robot Policies

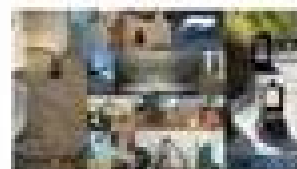
## Large Datasets

Leverage Existing Robot Data



## Large Models

Cross-embodied Policies



## Scalable Evaluation

Robot Evals are  
Tedious & Expensive

Difficult to reproduce

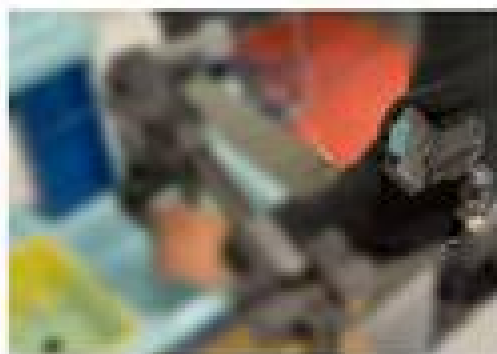
# Evaluating Policies is Expensive



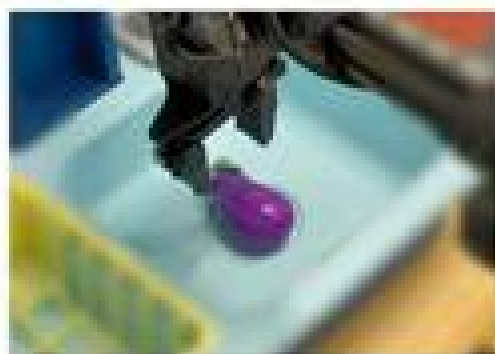
... people bump into cameras,

# Evaluating Policies is Expensive

- Sim evals democratize research
- Sim evals are more reproducible



... people bump into cameras,



grippers get stuck,



and real eval is slow and tedious.

# Evaluating Real-World Policies in Simulation

- Sim evals are **more reproducible**

... and make them **SIMPLER** real  
↓  
sim

Google Robot



BridgeData V2



Evaluating Real-World Robot Manipulation Policies in Simulation Policy

Li\*, Hsu\*, Gu\*, Pertsch†, Mees† et al., CoRL, 2024.

# Evaluating Real-World Policies in Simulation

- Sim evals are **more reproducible**

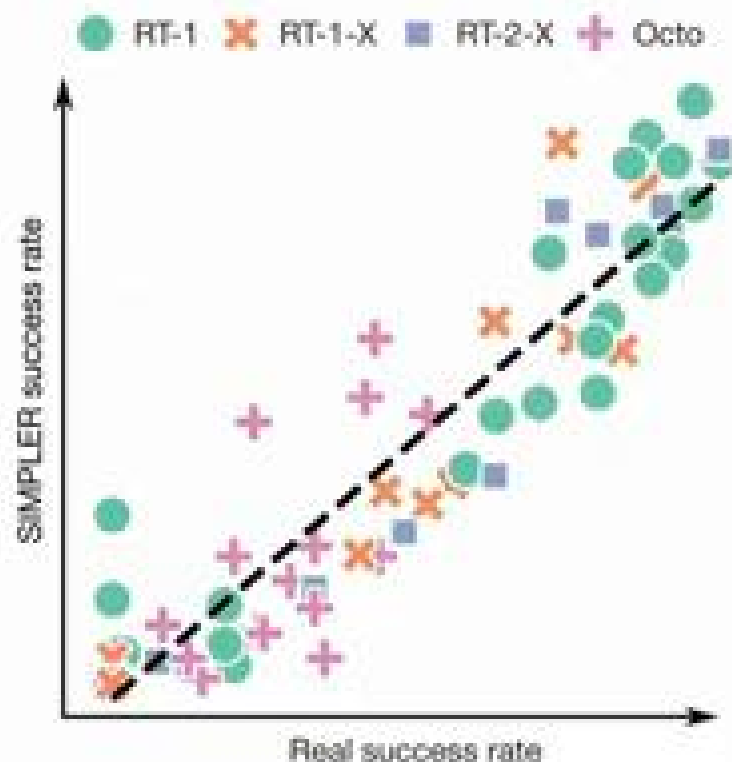


Evaluating Real-World Robot Manipulation Policies in Simulation Policy

Li\*, Hsu\*, Gu\*, Pertsch†, Meest† et al., CoRL, 2024.

# Evaluating Real-World Policies in Simulation

OK, but how meaningful are SIMPLER results?



How correlated are real and SIMPLER performance measures?

Very!

SIMPLER is a reliable proxy for real robot evaluation.

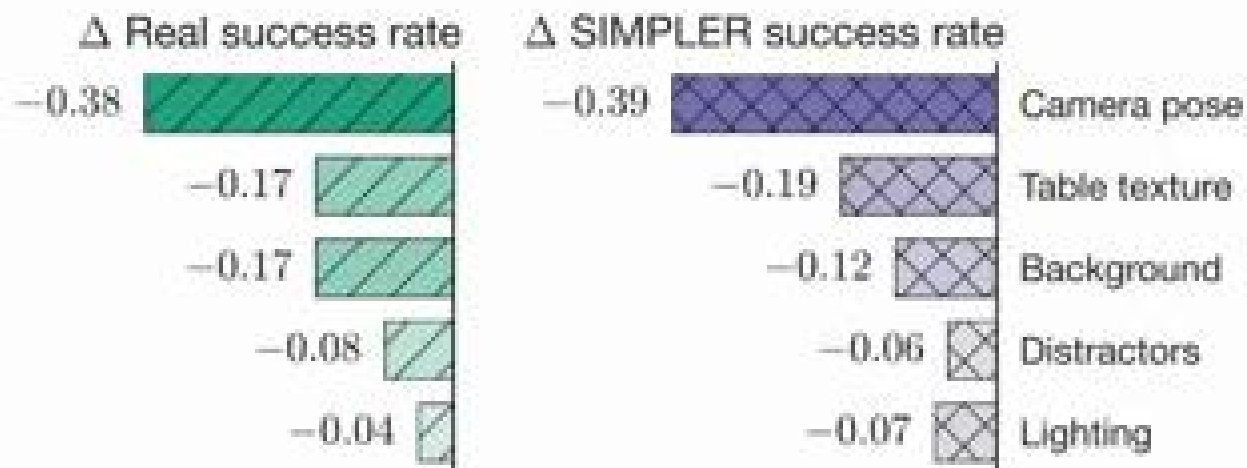
Evaluating Real-World Robot Manipulation Policies in Simulation Policy

Li\*, Hsu\*, Gu\*, Pertsch†, Mees† et al., CoRL, 2024.

# Evaluating Real-World Policies in Simulation

OK, but how meaningful are SIMPLER results?

SIMPLER can accurately characterize an RT-1 policy's robustness to various distribution shifts.



# Evaluating Real-World Policies in Simulation

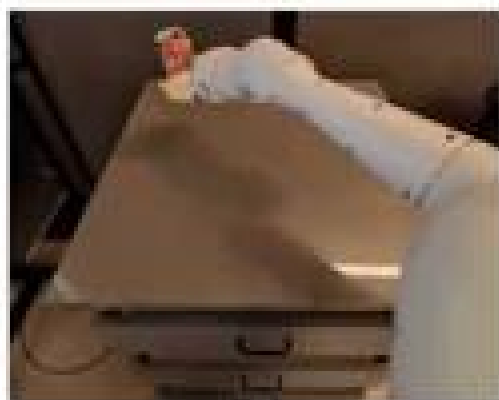
So, what matters for good simulated evaluation?

# Evaluating Real-World Policies in Simulation

So, what matters for good simulated evaluation?

1. Accurate control dynamics via system identification (SysID)

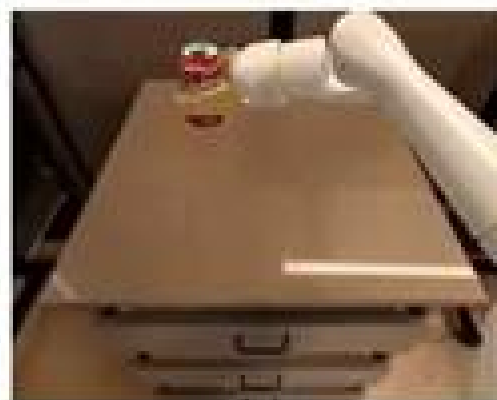
Same open-loop action sequence:



Real



Sim w/o SysID

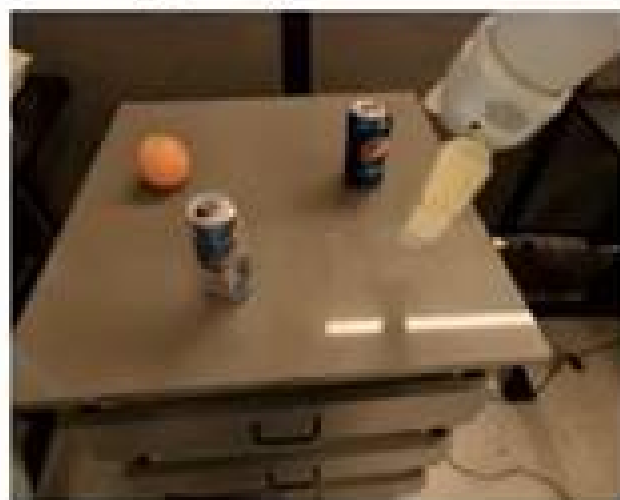


Sim after SysID (ours)

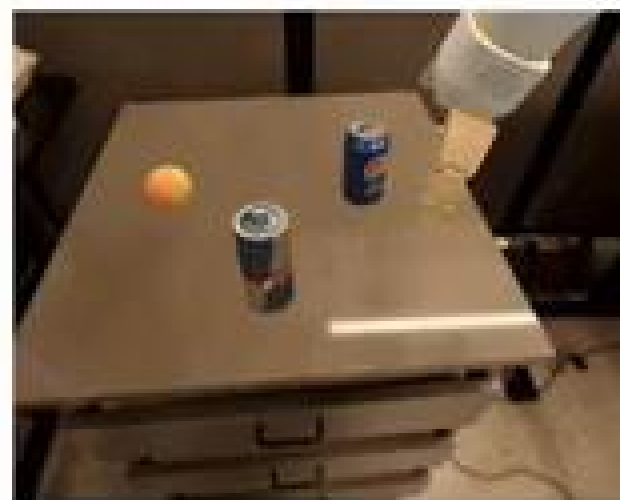
# Evaluating Real-World Policies in Simulation

So, what matters for good simulated evaluation?

2. Mitigating visual distribution shifts via "Visual Matching"



Real



Sim

+ Green Screen  
+ Texture Matching

Evaluating Real-World Robot Manipulation Policies in Simulation Policy

Li\*, Hsu\*, Gu\*, Pertsch†, Mees† et al., CoRL, 2024.

# Follow Up Works on Real-to-Sim Evals



**Polaris: Scalable Real-to-Sim Evaluations for Generalist Robot Policies**  
Jain et al., 2025.



**RobotArena : Scalable Robot Benchmarking via Real-to-Sim Translation**  
Jangir et al., 2025.



**RoboLab: A High-Fidelity Simulation Benchmark for Analysis of Task Generalist Policies**  
Yang et al., 2026.

# Ingredients for Generalist Robot Policies

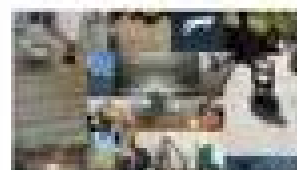
## Large Datasets

Leverage Existing Robot Data



## Large Models

Cross-embodied Policies with Internet-scale Priors



$\pi$

## Scalable Evaluation

Leverage Real2Sim Evals



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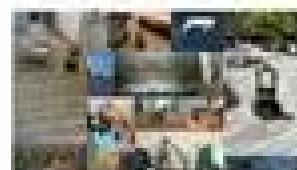
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**Thank you for your attention**